

Feedback — Programming Homework 3

[Help Center](#)

You submitted this homework on **Sat 25 Jul 2015 11:16 AM PDT**. You got a score of **4.00** out of **4.00**.

We saw how to adapt dynamic programming to find approximate occurrences of a pattern in a text. Recall that:

- Rows of the dynamic programming matrix are labeled with bases from P and columns with bases from T
- Elements in the first row are set to 0
- Elements in the first column are set to 0, 1, 2, ..., as for edit distance
- Other elements are set in the same way as elements of a standard edit distance matrix
- The minimal value in the bottom row is the edit distance of the closest match between P and T

First, download the provided excerpt of human chromosome 1

<https://d28rh4a8wq0iu5.cloudfront.net/ads1/data/chr1.GRCh38.excerpt.fasta>

Second, parse it using the `readGenome` function we wrote before.

Third, adapt the `editDistance` function we saw in practical (copied below) to solve this problem. Your function should take arguments `p` (pattern), `t` (text) and should return the edit distance of the match between P and T with the fewest edits.

```
def editDistance(x, y):
    # Create distance matrix
    D = []
    for i in range(len(x)+1):
        D.append([0]*(len(y)+1))

    # Initialize first row and column of matrix
    for i in range(len(x)+1):
        D[i][0] = i
    for i in range(len(y)+1):
        D[0][i] = i

    # Fill in the rest of the matrix
    for i in range(1, len(x)+1):
        for j in range(1, len(y)+1):
            distHor = D[i][j-1] + 1
            distVer = D[i-1][j] + 1
            if x[i-1] == y[j-1]:
                distDiag = D[i-1][j-1]
            else:
                distDiag = D[i-1][j-1] + 1
            D[i][j] = min(distHor, distVer, distDiag)
```

```
# Edit distance is the value in the bottom right corner of the matrix  
return D[-1][-1]
```

Question 1

What is the edit distance of the best match between pattern GCTGATCGATCGTACG and the excerpt of human chromosome 1?

You entered:

Your Answer		Score	Explanation
3	✓	1.00	
Total		1.00 / 1.00	

Question 2

What is the edit distance of the best match between pattern GATTTACCAGATTGAG and the excerpt of human chromosome 1?

You entered:

Your Answer		Score	Explanation
2	✓	1.00	
Total		1.00 / 1.00	

Question 3

In a practical, we saw a function for finding the longest exact overlap (suffix/prefix match)

between two strings. The function is copied below.

```
def overlap(a, b, min_length=3):
    """ Return length of longest suffix of 'a' matching
        a prefix of 'b' that is at least 'min_length'
        characters long.  If no such overlap exists,
        return 0. """
    start = 0 # start all the way at the left
    while True:
        start = a.find(b[:min_length], start) # look for b's suffix in a
        if start == -1: # no more occurrences to right
            return 0
        # found occurrence; check for full suffix/prefix match
        if b.startswith(a[start:]):
            return len(a)-start
        start += 1 # move just past previous match
```

Say we are only concerned with overlaps that (a) contain no differences, and (b) are at least k bases long. To make an overlap graph, we could call the overlap function on every pair of reads from the dataset, specifying `min_length=k`. Unfortunately, that will be very slow!

Let's use grouping to make construction of the overlap graph much faster. Let every possible k -mer have an associated set, which starts out empty. For every k -mer in a read, we add the read to the set corresponding to the k -mer. Once we have done this for every k -mer in every read, we do the following:

- For each read r :
 - For each read o in the set for k -mer $r[-k:]$ (i.e. the length- k suffix of r):
 - Check whether r overlaps o using `overlap(r, o, k)` and, if so, record the overlap

An important point is that we *do not* need to call `overlap` on a pair of reads if the the second read does not contain the length- k suffix of the first. So we avoid many unnecessary comparisons.

Download and parse the read sequences from the provided Phi-X FASTQ file. We'll just use their base sequences, so you can ignore read names and base qualities.

https://d28rh4a8wq0iu5.cloudfront.net/ads1/data/ERR266411_1.for_asm.fastq

Next, using the approach described above, find all the pairs of reads that have an exact suffix/prefix match of length at least 30. Don't compare reads to themselves. Note that looking for a suffix of A that matches a prefix of B is not the same as looking for a suffix of B that matches a prefix of A , so try both for every pair of reads.

- Hint 1: Your function should take **at most** 10 seconds to run on this 10,000-read dataset. If your function is much slower than this, there is probably an issue with it.
- Hint 2: The provided FASTQ file does not contain any duplicate reads

- Hint 3: **Remember not to overlap a read with itself.** If you do, your answers will be too high.

Picture the overlap graph corresponding to the overlaps just calculated. How many edges are in the graph? (In other words, how many distinct pairs of reads overlap?)

You entered:

904746

Your Answer		Score	Explanation
904746	✓	1.00	
Total		1.00 / 1.00	

Question 4

Picture the overlap graph corresponding to the overlaps computed for the previous question. How many nodes in this graph have at least one outgoing edge? (In other words, how many reads have a suffix involved in an overlap?)

You entered:

7161

Your Answer		Score	Explanation
7161	✓	1.00	
Total		1.00 / 1.00	